

I. Problem

Construct a Deterministic Finite Automata (DFA) that recognizes a single C-style comment, and implement it in C. Using the DFA, determine whether each of the following input strings is accepted or rejected:

1. /*aaa*aaa*/
2. /*aaa*aaa*/a

II. Given

$$\Sigma = \{a, *, /\}$$

III. Solution

$M = (Q, \Sigma, \delta, q_0, F)$ where

$$Q = \{Q_0, Q_1, Q_2, Q_3, Q_4, DEAD\}$$

$$\Sigma = \{a, *, /\}$$

$$q_0 = Q_0$$

$$F = \{Q_4\}$$

$$\Delta: Q \times \Sigma \rightarrow Q$$

δ	a	*	/
$\rightarrow Q_0$	DEAD	DEAD	Q_1
Q_1	DEAD	Q_2	DEAD
Q_2	Q_2	Q_3	Q_2
Q_3	Q_2	Q_3	Q_4
$*Q_4$	DEAD	DEAD	DEAD
DEAD	DEAD	DEAD	DEAD

